

Project Executable Files

Date	30 June 2026
Team ID	
Project Name	Rising Waters: A Machine Learning Approach to Flood Prediction
Maximum Marks	3 Marks

Project Executable Files

This section contains all details regarding the project executables, organization, run instructions, and verification checklists.

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide (instructions to run the project)	Yes
3	requirements.txt / package.json / dependency file	Yes
4	Database schema / seed files (if applicable)	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL (if hosted)	Yes
7	APK / Executable binary (if applicable for mobile/desktop apps)	NA
8	Dockerfile / Containerization config (if applicable)	NA
9	Test files and test results	Yes
10	Demo video or walkthrough (if required)	Yes

Step 2: File / Folder Structure

Provide an overview of your project's file and folder structure below:

Flood-Prediction-main/

```
|
|  └─ app.py                # Main Flask application
|  └─ db_setup.py           # Database initialization
|  └─ preprocess.py         # Data preprocessing
|  └─ train.py              # ML model training script
|  └─ requirements.txt       # Python dependencies
|  └─ readme.md             # Project documentation
|  └─ .gitignore            # Git ignore configuration
|
|  └─ data/
|    └─ flood_prediction.db  # SQLite database
|
|  └─ models/
|    └─ flood_model.pkl      # Trained ML model
|    └─ scaler.save          # Saved scaler
|    └─ model_meta.json      # Model metadata
|    └─ feature_importance.png # Feature importance graph
|    └─ roc_curve.png        # ROC curve
|
|  └─ static/
|    └─ css/
|      └─ main.css           # Stylesheet
|    └─ js/
|      └─ main.js            # JavaScript
|    └─ feature_importance.png
|    └─ roc_curve.png
|
└─ templates/
  └─ index.html             # Landing page
  └─ home.html              # Home page
  └─ login.html             # Login page
  └─ register.html          # Registration page
  └─ dashboard.html         # User dashboard
  └─ chance.html            # Flood prediction result
```

```

└─ no_chance.html      # No flood prediction result
└─ prediction_history.html # Prediction history
└─ profile.html        # User profile
└─ manage_users.html    # Admin user management
└─ model_dashboard.html # Model analytics dashboard
  
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://flood-prediction-4rwg.onrender.com/ (Also local: http://127.0.0.1:5001)
Login Credentials (Demo)	Email: admin@flood.com Password: admin
Platform / Hosting Provider	Render
Repository Link	https://github.com/sameer727/Flood-Prediction
Demo Video Link	https://drive.google.com/file/d/1nhkH5yJtJRYzrYU5z_xhBQNljjEndlDG/view?usp=sharing

Step 4: Run Instructions

Provide clear, step-by-step instructions for the evaluator to run your project locally:

Pre-requisites:

- **Python:** version 3.12 or 3.13 installed.
- **Git:** installed (for cloning).
- **Web Browser:** modern browser (Chrome, Edge, Firefox).

Step 1: Clone the repository

Open a shell terminal and run:

```
git clone https://github.com/sameer727/Flood-Prediction.git
cd Flood-Prediction
```

Step 2: Set up Virtual Environment & Install Dependencies

Initialize a clean Python virtual environment and run package installation:

```
# Initialize virtual environment
python -m venv .venv
```

```
# Activate the virtual environment
```

```
# On Windows:
```

```
.venv\Scripts\activate
```

```
# On macOS/Linux:
```

```
source .venv/bin/activate
```

```
# Install required packages
```

```
pip install -r requirements.txt
```

Step 3: Run Database Setup & Seeds

Execute the schema creation script to create tables and set up the default admin user:

```
python db_setup.py
```

Step 4: Tune Database Concurrency

Execute the optimization script to configure WAL mode and index foreign keys:

```
python optimize_db.py
```

Step 5: Start the Application

Boot the web server locally using Python:

```
python app.py
```

Step 6: Open Web Browser

Open your browser and navigate to:

```
http://127.0.0.1:5001
```

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	SQLite Concurrent locking: Under high virtual user write load, standard SQLite rollback journaling locks the database and queues requests.	Resolved: Write-Ahead Logging (WAL) mode enabled and PRAGMA synchronous = NORMAL configured to allow parallel reads and writes.
2	Single-threaded dev server limits: Running heavy joins (history query logs) under high concurrency can create queue delays because the Werkzeug dev server is single-threaded.	Status: Local development behaves as expected. For production deployment, using a multi-worker WSGI server like Gunicorn or uWSGI is recommended.

3	Bcrypt authentication CPU load: High concurrency logins can saturate the CPU due to the resource-intensive hashing computation.	Status: Session persistence cookies are used in performance tests to bypass redundant hashing executions.
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